AI-Enhanced Intrusion Detection System with Intelligent Risk Assessment

Senior Capstone Project | Computer Engineering | Graduation 2025

Cybersecurity Machine Learning Raspberry Pi Real-Time Monitoring

97%+	5	Real-Time	Low Cost
Detection Accuracy	Threat Levels	Network Monitoring	Raspberry Pi 5

How It Works

01 Network Traffic — *Data enters the system*

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02 Snort IDS Engine — *Detects suspicious packets*

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03 AI Model — *Classifies: Normal / Low / Medium / High / Critical*

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04 Risk Scoring — *Prioritizes alerts by severity*

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05 Dashboard — *Admin sees results in real time*

Project Overview

! The Problem

- Traditional firewalls miss complex modern threats
- Manual alert triage is slow and error-prone
- No automatic priority for critical vs minor threats
- Expensive enterprise IDS out of reach for small orgs

✓ Our Solution

- Snort IDS on Raspberry Pi 5 captures all network traffic
- AI (Random Forest) classifies each alert automatically
- Risk scoring: Critical > High > Medium > Low > Normal
- Live dashboard shows alerts and sends notifications

★ Why It Matters

- Affordable hardware (~$80) replaces costly systems
- 97%+ accuracy using CIC-IDS 2017 benchmark dataset
- Reduces false alarms that waste analyst time
- Scalable to homes, small businesses, and universities

Technology Used

Hardware	Software	AI Model	Threat Levels
Raspberry Pi 5 MicroSD 256GB Network Interface Power Supply 5V/3A	Snort IDS Python 3 Scikit-Learn Flask Dashboard	Random Forest CIC-IDS 2017 dataset 97%+ accuracy Real-time inference	**Critical / High / Medium / Low / Normal**

Project Team

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